

IOQM Problem Set

Number Theory

Shounak Kar

Here are some practice problems in Number theory for IOQM. The answer to each question is a positive integer (ranging from 1 to 999).

Problem 1

Let $abba$ be a four-digit palindrome that is divisible by b and is also divisible by the two-digit number bb . Find the number of palindromes satisfying both of these properties.

Problem 2

A positive integer n is called *bad* if it cannot be expressed as the product of two distinct positive integers greater than 1. Find the number of *bad* positive integers less than 100.

Problem 3

Let S be the set of integers of the form $2^a + 2^b + 2^c$, where a, b, c are pairwise distinct non-negative integers. Determine the 100th smallest element of S .

Problem 4

Find the sum of all positive integers n such that $\sqrt{n^2 + 85n + 2017}$ is an integer.

Problem 5

A point whose coordinates are both integers is called a lattice point. How many lattice points lie on the hyperbola $x^2 - y^2 = 2000^2$?

Problem 6

Suppose that x , y , and z are three positive numbers that satisfy the equations $xyz = 1$, $x + \frac{1}{z} = 5$, and $y + \frac{1}{x} = 29$. Then $z + \frac{1}{y} = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Problem 7

A set of positive numbers has the triangle property if it has three distinct elements that are the lengths of the sides of a triangle whose area is positive. Consider sets $\{4, 5, 6, \dots, n\}$ of consecutive positive integers, all of whose ten-element subsets have the triangle property. What is the largest possible value of n ?

Problem 8

How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where i and j are integers and $0 \leq i < j \leq 99$?

Problem 9

A set $\mathcal{S}$ of distinct positive integers has the following property: for every integer x in $\mathcal{S}$, the arithmetic mean of the set of values obtained by deleting x from $\mathcal{S}$ is an integer. Given that 1 belongs to $\mathcal{S}$ and that 2002 is the largest element of $\mathcal{S}$, what is the greatest number of elements that $\mathcal{S}$ can have?

Problem 10

Find the least positive integer k for which the equation $\left\lfloor \frac{2002}{n} \right\rfloor = k$ has no integer solutions for n . (The notation $\lfloor x \rfloor$ means the greatest integer less than or equal to x .)

Problem 11

Let the set $\mathcal{S} = \{8, 5, 1, 13, 34, 3, 21, 2\}$. Ahan makes a list as follows: for each two-element subset of $\mathcal{S}$, he writes on her list the greater of the set's two elements. Find the sum of the numbers on the list.

Problem 12

Two positive integers differ by 60. The sum of their square roots is the square root of an integer that is not a perfect square. What is the maximum possible sum of the two integers?

Problem 13

How many positive integers less than 10,000 have at most two different digits?

Problem 14

The number n can be written in base 14 as $\underline{a} \underline{b} \underline{c}$, can be written in base 15 as $\underline{a} \underline{c} \underline{b}$, and can be written in base 6 as $\underline{a} \underline{c} \underline{a} \underline{c}$, where $a > 0$. Find the base-10 representation of n .

Problem 15

Find the least positive integer n such that when 3^n is written in base 143, its two right-most digits in base 143 are 01.

Problem 16

For a positive integer n , let d_n be the units digit of $1 + 2 + \cdots + n$. Find the remainder when

$$\sum_{n=1}^{2017} d_n$$

is divided by 1000.

Problem 17

Find the number of sets $\{a, b, c\}$ of three distinct positive integers with the property that the product of a, b , and c is equal to the product of 11, 21, 31, 41, 51, 61.

Problem 18

Compute the sum of all 2-digit prime numbers p such that there exists a prime number q for which $100q + p$ is a perfect square.

Problem 19

Compute the sum of all positive integers n for which the expression

$$\frac{n+7}{\sqrt{n-1}}$$

is an integer.

Problem 20

What is the smallest positive integer that cannot be written as the sum of two non-negative palindromic integers? (An integer is palindromic if the sequence of decimal digits are the same when read backwards.)

Problem 21

There are two prime numbers p so that $5p$ can be expressed in the form $\left\lfloor \frac{n^2}{5} \right\rfloor$ for some positive integer n . What is the sum of these two prime numbers?

Problem 22

Suppose that a and b are real numbers such that the line $y = ax + b$ intersects the graph of $y = x^2$ at two distinct points A and B . If the coordinates of the midpoint of AB are $(5, 101)$, compute $a + b$.

Problem 23

Find the sum of all prime numbers p such that p divides

$$(p^2 + p + 20)^{p^2+p+2} + 4(p^2 + p + 22)^{p^2-p+4}.$$

Problem 24

Compute the sum of all positive integers whose positive divisors sum to 186.

Problem 25

Compute the number of non-negative integral ordered pairs (x, y) such that $x^2 + y^2 = 32045$.

Problem 26

The smallest three positive proper divisors of an integer n are $d_1 < d_2 < d_3$ so that $d_1 + d_2 + d_3 = 57$. Find the sum of the possible values of d_2 .

Problem 27

Compute the remainder when $2^{3^5} + 3^{5^2} + 5^{2^3}$ is divided by 30.

Problem 28

How many ordered triples of non-zero integers (a, b, c) satisfy $2abc = a + b + c + 4$?

Problem 29

Find the least positive integer N such that the set of 1000 consecutive integers beginning with $1000 \cdot N$ contains no square of an integer.

Problem 30

The positive integers N and N^2 both end in the same sequence of four digits $abcd$ when written in base 10, where digit a is not zero. Find the three-digit number abc .

Problem 31

There is a prime number p such that $16p + 1$ is the cube of a positive integer. Find p .

Problem 32

For positive integer n , let $s(n)$ denote the sum of the digits of n . Find the smallest positive integer satisfying $s(n) = s(n + 864) = 20$.

Problem 33

For how many integers n between 1 and 50, inclusive, is

$$\frac{(n^2 - 1)!}{(n!)^n}$$

an integer? (Recall that $0! = 1$.)

Problem 34

Let S be the set of all positive integer divisors of 100,000. How many numbers are the product of two distinct elements of S ?

Problem 35

Find the least positive integer such that when its leftmost digit is deleted, the resulting integer is $\frac{1}{29}$ of the original integer.

Problem 36

Let P be the product of the first 100 positive odd integers. Find the largest integer k such that P is divisible by 3^k .

Problem 37

How many integers N less than 1000 can be written as the sum of j consecutive positive odd integers from exactly 5 values of $j \geq 1$?

Problem 38

Let S_n be the sum of the reciprocals of the non-zero digits of the integers from 1 to 10^n inclusive. Find the smallest positive integer n for which S_n is an integer.

Problem 39

For each positive integer p , let $b(p)$ denote the unique positive integer k such that $|k - \sqrt{p}| < \frac{1}{2}$. For example, $b(6) = 2$ and $b(23) = 5$. If $S = \sum_{p=1}^{2007} b(p)$, find the remainder when S is divided by 1000.

Problem 40

Distinct prime numbers p, q, r satisfy the equation

$$2pqr + 50pq = 7pqr + 55pr = 8pqr + 12qr = A$$

for some positive integer A . What is sum of digits of A ?

Problem 41

Compute the number of positive integers $n \leq 1000$ such that $\text{lcm}(n, 9)$ is a perfect square. (Recall that lcm denotes the least common multiple.)

Problem 42

Compute the sum of all positive integers n such that $50 \leq n \leq 100$ and $2n + 3$ does not divide $2^{n!} - 1$.

Problem 43

Find the number of positive integers $n < 2018$ such that $25^n + 9^n$ is divisible by 13.

Problem 44

For a positive integer n , let $f(n)$ be the number of (not necessarily distinct) primes in the prime factorization of n . For example, $f(1) = 0$, $f(2) = 1$, and $f(4) = f(6) = 2$. Let $g(n)$ be the number of positive integers $k \leq n$ such that $f(k) \geq f(j)$ for all $j \leq n$. Find $g(1) + g(2) + \cdots + g(100)$.

Problem 45

The sequence of positive integers $a_1, a_2, \dots$ has the property that $\gcd(a_m, a_n) > 1$ if and only if $|m - n| = 1$. Find the sum of the four smallest possible values of a_2 .

Problem 46

Compute the smallest positive integer n such that there do not exist integers x and y satisfying $n = x^3 + 3y^3$.

Problem 47

Compute the sum of all positive integers n such that the median of the n smallest prime numbers is n .

Problem 48

Given a positive integer n , let $p(n)$ be the product of the non-zero digits of n . (If n has only one digit, then $p(n)$ is equal to that digit.) Let $S = p(1) + p(2) + p(3) + \cdots + p(999)$. What is the largest prime factor of S ?

Problem 49

Find the number of positive integers m for which there exist non-negative integers $x_0, x_1, \dots, x_{2011}$ such that

$$m^{x_0} = \sum_{k=1}^{2011} m^{x_k}.$$

Problem 50

Set $S = \{105, 106, \dots, 210\}$. Determine the minimum value of n such that any n -element subset T of S contains at least two non-relatively prime elements.

Answer Key

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| 1. 31 | 18. 179 | 35. 725 |
| 2. 30 | 19. 89 | 36. 49 |
| 3. 577 | 20. 21 | 37. 15 |
| 4. 195 | 21. 52 | 38. 63 |
| 5. 98 | 22. 61 | 39. 955 |
| 6. 5 | 23. 344 | 40. 18 |
| 7. 253 | 24. 202 | 41. 43 |
| 8. 784 | 25. 16 | 42. 222 |
| 9. 30 | 26. 42 | 43. 336 |
| 10. 49 | 27. 6 | 44. 136 |
| 11. 484 | 28. 6 | 45. 42 |
| 12. 156 | 29. 282 | 46. 6 |
| 13. 927 | 30. 937 | 47. 25 |
| 14. 925 | 31. 307 | 48. 103 |
| 15. 195 | 32. 695 | 49. 16 |
| 16. 69 | 33. 34 | 50. 26 |
| 17. 728 | 34. 117 | |